Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer pursuing an MSCS at USC with 2+ years building **simulation and benchmarking pipelines**, **performance-optimized systems**, and concurrent tooling in **C++ and Python**, with strong **computer architecture**, algorithms, and ML-workload foundations.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Investigated and implemented **performance optimizations** across backend services for 500+ users, reducing request latency from 1.2s to 350ms by profiling hot paths, tuning execution, and optimizing data-access patterns.
- Built **fault-tolerant async orchestration and tooling** for 5K+ monthly jobs with explicit state transitions and failure isolation, improving throughput **3x** and enabling efficient debug and correlation across concurrent workloads.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered a **concurrent processing engine** with multithreading, benchmarking throughput across 1K+ files per batch and 10K+ daily requests while optimizing execution paths for stable performance under sustained load.
- Built **backend tooling and instrumentation** tracking per-operation cost and usage, enabling data-driven performance and tradeoff decisions across high-volume model-execution workflows.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **simulation and benchmarking pipelines** over 100K+ records for closed-loop evaluation, enabling reproducible performance correlation and high-confidence debugging of model-driven decision systems and architectures.
- Architected a **transformer-based model** with multi-head attention for ML workloads, improving long-horizon planning **18%** through profiling, performance-focused experimentation, and rigorous evaluation across simulation rollouts.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed **test automation tools and infrastructure** (nightly VM builds, 20+ unit/integration suites) with Jenkins, PyTest, and Mockito across 3 OSes, increasing CI reliability **68%** and accelerating debug cycles.
- Built **object-oriented modules** (Singleton, Factory) for large-scale integrations with AWS Glue and S3, optimizing algorithms and improving response times **30%** across distributed, cross-platform production workloads.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **real-time microservice and WebSocket modules** syncing 2M+ records, optimizing throughput consistency and low-latency responsiveness for high-volume concurrent processes across distributed services.
- Automated **Docker-based build and test pipelines** with Jenkins, cutting release time **50%**, improving deployment consistency, and reducing environment drift that previously slowed validation and troubleshooting.

Skills

Languages: C++, Python, Java, JavaScript/TypeScript, SQL, Bash
Systems & Architecture: Computer Architecture, Operating Systems, Concurrency, Multithreading, Data Structures, Algorithms
Simulation & Performance: Simulators, Performance Modeling, Benchmarking, Profiling, Performance Optimization, Closed-Loop Evaluation
ML & Accelerators: PyTorch, Transformers, ML Workloads, GPU/Accelerator-Aware Optimization, Hyperparameter Tuning
Tooling & Infrastructure: Test Automation, CI/CD, Jenkins, Docker, Git/GitHub, Linux
Backend: REST APIs, FastAPI, Flask, Microservices, Distributed Systems

Projects

CPU-Scheduling Simulator Live | *C++, Simulation, OS Internals, Algorithms*
- Built a **C++ CPU-scheduling simulator** modeling FCFS, SJF, SRT, Round Robin, and multi-level queue algorithms with Gantt visualization, reducing average turnaround **25%** across 1K+ simulated process configurations.

Big-Brain Live | *Cloud Pipeline, Performance Tuning*
- Built a **cloud data pipeline** indexing 50K+ records with incremental updates and performance tuning, reducing retrieval latency **40%** through optimized embedding and similarity-search execution paths.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Built a **containerized REST processing API** on AWS (CodePipeline, ECS/Fargate) with automated CI/CD, validation gates, and reliability tooling, streamlining testing and debug across production workflows.